



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

BCSE204L- Design and Analysis of Algorithms

Dr. Iyappan Perumal

Assistant Professor Senior Grade 2

School of Computer Science & Engineering

VIT,Vellore.



Course Objective

1. To provide mathematical foundations for analysing the complexity of the algorithms
2. To impart the knowledge on various design strategies that can help in solving the real world problems effectively
3. To synthesize efficient algorithms in various engineering design situations



BCSE204L- Design and Analysis of Algorithms-

Course outcome

1. Apply the mathematical tools to analyse and derive the running time of the algorithms
2. Demonstrate the major algorithm design paradigms.
3. Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
4. Articulating Randomized Algorithms.
5. Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.



BCSE204L- Design and Analysis of Algorithms- Syllabus

Module:1	Design Paradigms: Greedy, Divide and Conquer Techniques	6 hours
Overview and Importance of Algorithms - Stages of algorithm development: Describing the problem, Identifying a suitable technique, Design of an algorithm, Derive Time Complexity, Proof of Correctness of the algorithm, Illustration of Design Stages - Greedy techniques: Fractional Knapsack Problem, and Huffman coding - Divide and Conquer: Maximum Subarray, Karatsuba faster integer multiplication algorithm.		
Module:2	Design Paradigms: Dynamic Programming, Backtracking and Branch & Bound Techniques	10 hours
Dynamic programming: Assembly Line Scheduling, Matrix Chain Multiplication, Longest Common Subsequence, 0-1 Knapsack, TSP- Backtracking: N-Queens problem, Subset Sum, Graph Coloring- Branch & Bound: LIFO-BB and FIFO BB methods: Job Selection problem, 0-1 Knapsack Problem		
Module:3	String Matching Algorithms	5 hours
Naïve String-matching Algorithms, KMP algorithm, Rabin-Karp Algorithm, Suffix Trees.		
Module:4	Graph Algorithms	6 hours
All pair shortest path: Bellman Ford Algorithm, Floyd-Warshall Algorithm - Network Flows: Flow Networks, Maximum Flows: Ford-Fulkerson, Edmond-Karp, Push Re-label Algorithm – Application of Max Flow to maximum matching problem		
Module:5	Geometric Algorithms	4 hours
<u>Line Segments: Properties, Intersection, sweeping lines</u> - Convex Hull finding algorithms: Graham's Scan, Jarvis' March Algorithm.		
Module:6	Randomized algorithms	5 hours
Randomized quick sort - The hiring problem - Finding the global Minimum Cut.		
Module:7	Classes of Complexity and Approximation Algorithms	7 hours
The Class P - The Class NP - Reducibility and NP-completeness – SAT (Problem Definition and statement), 3SAT, Independent Set, Clique, Approximation Algorithm – Vertex Cover, Set Cover and Travelling salesman		
Module:8	Contemporary Issues	2 hours



BCSE204L- Design and Analysis of Algorithms- References

Text Books:

1. Thomas H. Cormen, C.E. Leiserson, R. L. Rivest and C. Stein, Introduction to Algorithms, 2009, 3rd Edition, MIT Press.

Reference Books:

1. Jon Kleinberg, Éva Tardos, Algorithm Design, Pearson education, 2014.
2. Rajeev Motwani, Prabhakar Raghavan; Randomized Algorithms, Cambridge University Press, 1995 (Online Print – 2013)
3. Ravindra K. Ahuja, Thomas L. Magnanti and James B. Orlin, “ Network Flows: Theory, Algorithms and Applications”, Pearson Education, 2014.



Module 5 : Geometric Algorithms (4 Hours)

- **Line Segments, Properties, Intersection, Sweeping Lines**
- Convex hull finding Algorithm
- Graham's Scan
- Jarvis March Algorithm



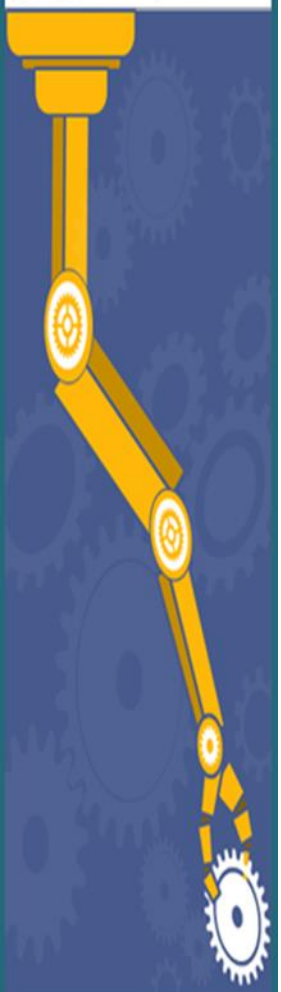
Computational Geometry

- Branch of computer science that studies algorithms for solving geometric problems.
- Used in various fields such as computer graphics, robotics, VLSI design, computer-aided design, molecular modelling, metallurgy, manufacturing, Textile layout, forestry and statistics.
- Input to Computational Geometry Problem
 - Set of geometric objects such as a **set of points**, a **set of line segments**, or the **vertices of a polygon** in counter clock- wise order.
- Output to Computational Geometry Problem
 - Response to a query about the objects such as **whether any of the lines intersect**, or perhaps a **new geometric object**, such as the **convex hull** (smallest enclosing convex polygon) of the set of points.



Computational Geometry

- Here Problems are represented in 2D- i.e in the plane
- Represents each input object by a set of points $\{p_1, p_2, p_3, \dots\}$ where each $p_i = (x_i, y_i)$ and x_i, y_i belongs to Real numbers.
- For example, we represent an **n-vertex polygon P** by a sequence $(p_0, p_1, p_2, \dots, p_{n-1})$ of its vertices in order of their appearance on the boundary of P.
- *Note: Able to apply in 3D as well as higher dimensional spaces, such problems and their solutions can be very difficult to visualize.*

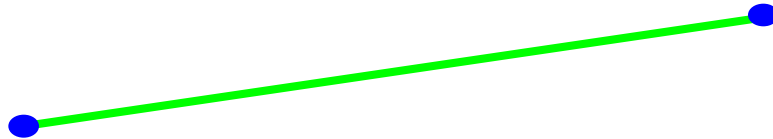


Basic Geometric Objects in the Plane

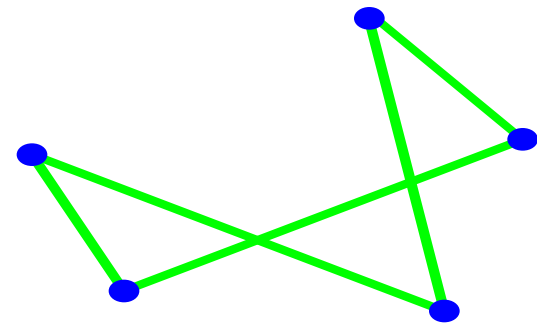
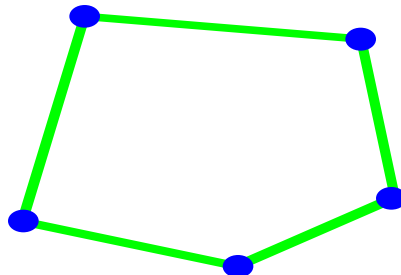
Point : denoted by a pair of coordinates (x,y)



Segment : Portion of a straight line between two points



Polygon : circular sequence of points (**vertices**) and segments (**edges**)



Line Segments

- Basic Questions about Line segments and how to answer??
- Whether one segment is **clockwise** or **counter clockwise** from another that shares an endpoint, **which way we turn** when traversing two adjoining line segments, and whether two line segments intersect??
 - Sweeping – to determine whether set of line segments contain any intersections.
 - Rotational Sweep-Computation of Convex hull (Smallest enclosing convex polygon of set of n points)
 - Graham's Scan & Jarvis March



Line and Line Segments

Line	Line Segment
No definite Endpoints	Definite Endpoints
Extended in both the directions	Can Extend but not in both the direction
Length of the line is indefinite	Length of the line is definite
Has no start or end point	Has both start and end point

Convex Combination : A convex combination of two distinct points $p_1(x_1, y_1)$ and $p_2(x_2, y_2)$ is any point $p_3(x_3, y_3)$ such that for some α in the range $0 \leq \alpha \leq 1$, we have $x_3 = \alpha x_1 + (1 - \alpha)x_2$ and $y_3 = \alpha y_1 + (1 - \alpha)y_2$. We also write that $p_3 = \alpha p_1 + (1 - \alpha)p_2$.



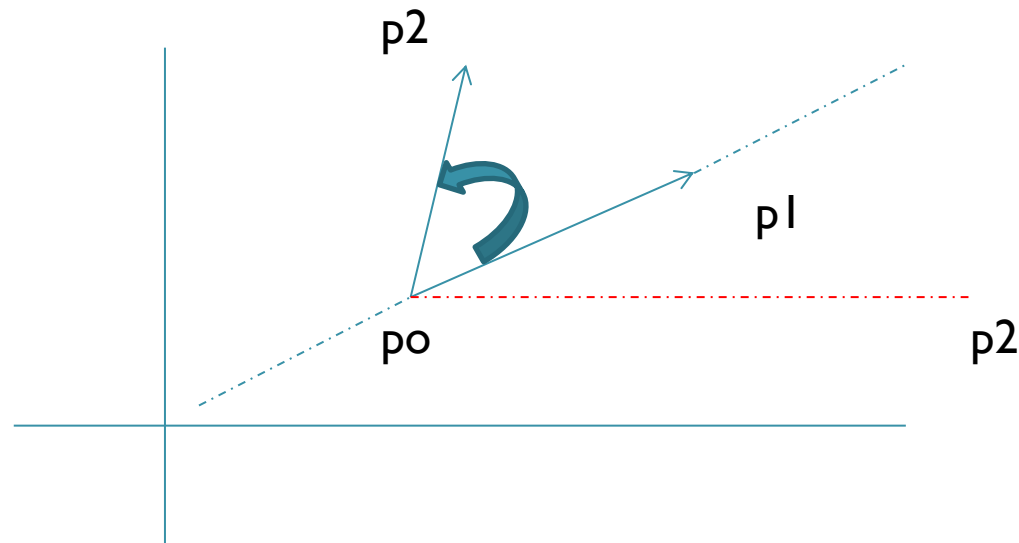


Line Segments- observation

- P3 is any point that is on the **line passing through p1 and p2** and is on or **between p1 and p2 on the line**.
- Given two distinct points p1 and p2 , the line segment p1p2 is the set of convex combinations of p1 and p2.
- We call p1 and p2 are the endpoints of segment p1 p2
- Sometimes the ordering of p1 and p2 matters, and we speak of the directed **segment $\overrightarrow{p1p2}$** . If p1 is the origin(0,0), then we can treat the directed segment **p1p2 as the vector p2** .

Questions related to Line Segments

- Given two **directed segments** $\overrightarrow{p_0p_1}$ and $\overrightarrow{p_0p_2}$, is $\overrightarrow{p_0p_1}$ clockwise from $\overrightarrow{p_0p_2}$ with respect to their common endpoint p_0 ?



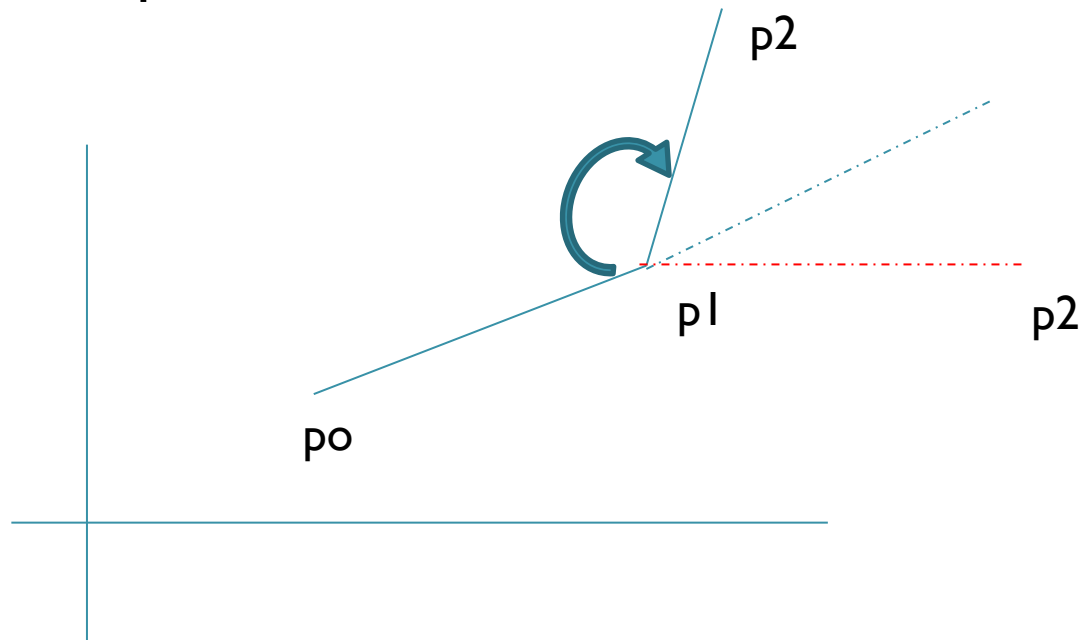
If I have to reach p_0p_2 from p_0p_1 which is nearer-clockwise or counter clockwise

P_0p_2 is counter clockwise from p_0p_1 - left turn at p_1



Questions related to Line Segments

- Given two **line segments** $\overline{p_0p_1}$ and $\overline{p_1p_2}$, if we traverse $\overline{p_0p_1}$ and then $\overline{p_1p_2}$, do we make left or right turn at point p_1 ??

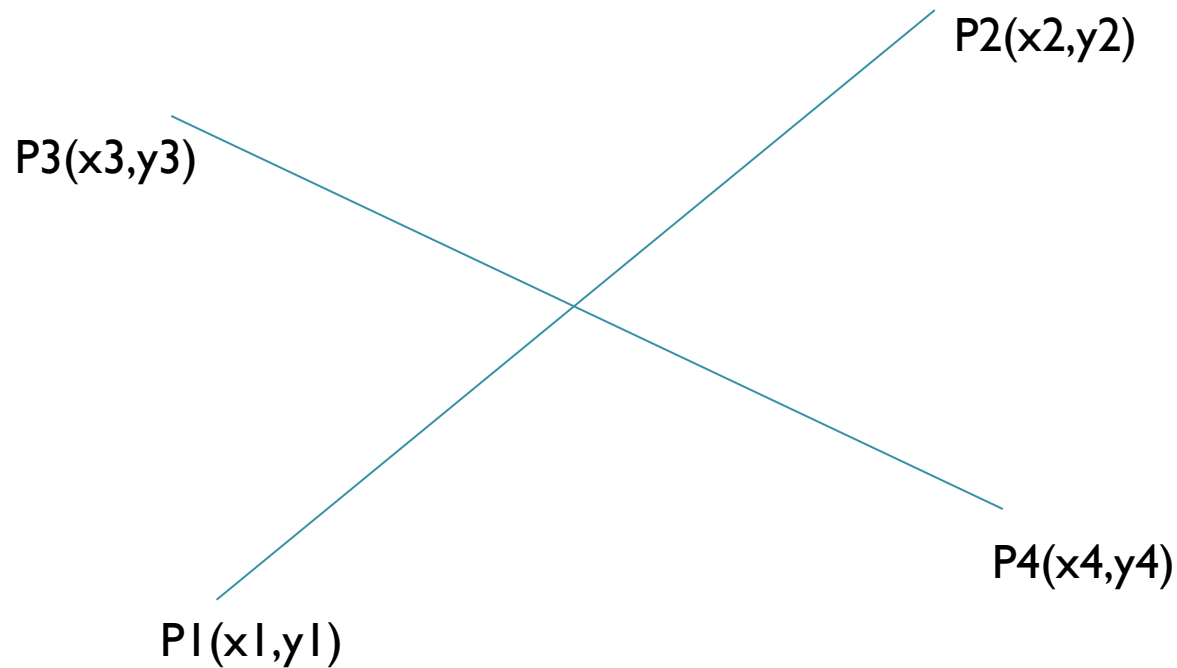


For above case: – Left Turn



Questions related to Line Segments

- Do **line segments p_1p_2 and p_3p_4** intersect??

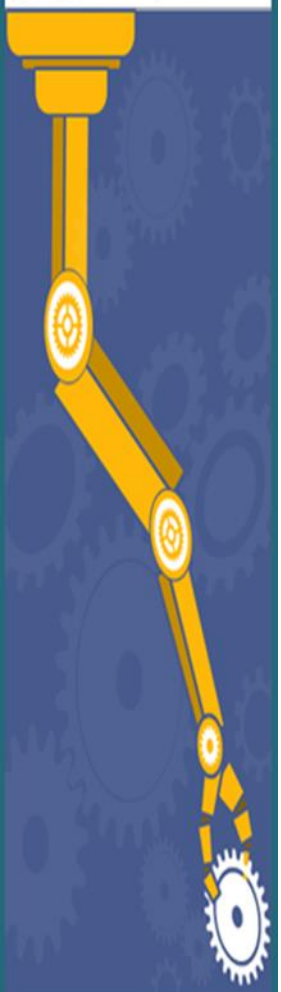


Note: There are no restrictions on the given points

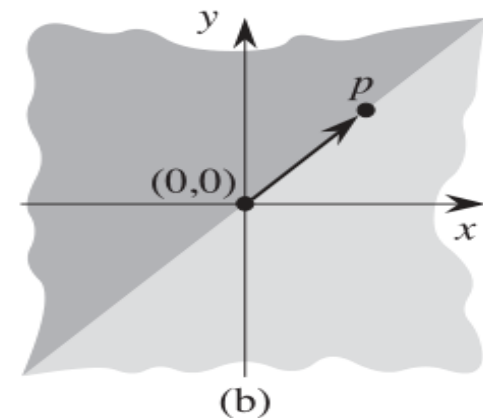
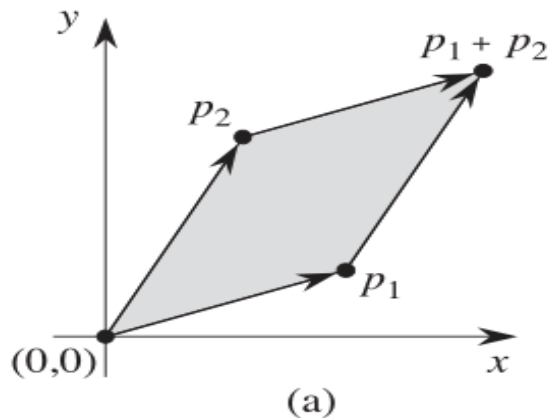


Line Segments Properties

- To handle above Questions some tools are used:
- **Tool I: Vector representation**
- A vector $\overrightarrow{p_0p_1}$ with $p_0 = (0,0)$, can be written as $\overrightarrow{p_1}$.



Line Segments Properties- cross products



Cross product $p_1 * p_2$ gives as the signed area of the parallelogram formed by the points $0,0$, p_1 , p_2 , and $p_1 + p_2 = (x_1 + x_2, y_1 + y_2)$.

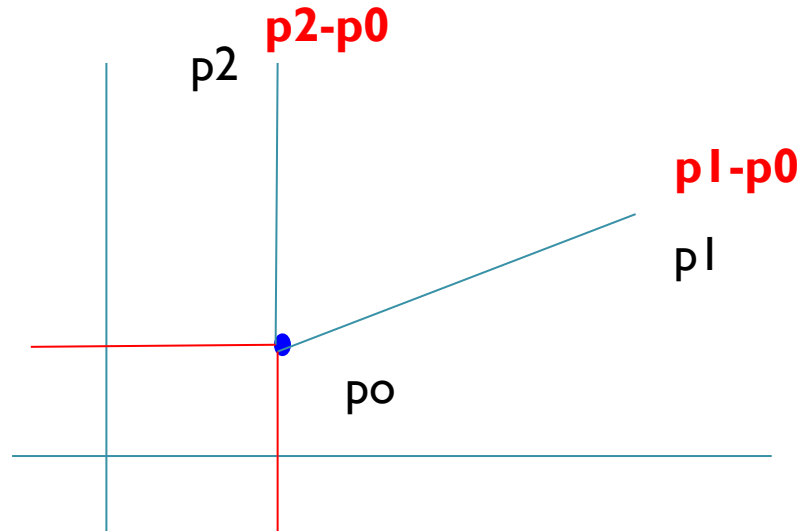
$$\vec{p_1} * \vec{p_2} = \text{Det} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}$$

$$= x_1 y_2 - y_1 x_2$$

$$= -(p_2 * p_1)$$

- If $\overline{p_1 * p_2} = +ve$, p_1 is clockwise from p_2 with respect to origin.
- If $\overline{p_1 * p_2} < 0$, p_1 is counter clockwise from p_2
- If 0 , its colinear, pointing in either same or opposite direction

Line Segments Properties- cross products



We let consider $p1 - p0$ denote the vector $\mathbf{p1'} = (x1', y1')$ where $x1' = x1 - x0$ and $y1' = y1 - y0$ and we define $p2 - p0$ similarly as $\mathbf{p2'} = (x2', y2')$.

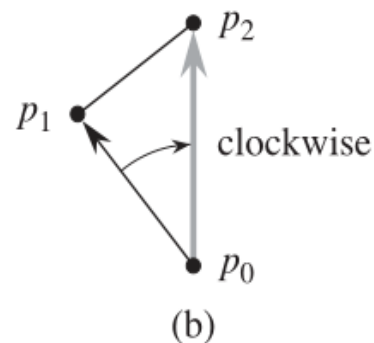
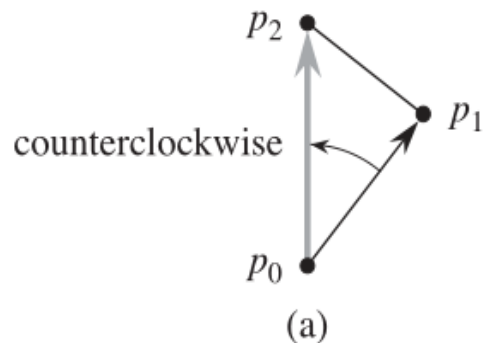
Now Cross product is calculated as:

$$\mathbf{p1'} * \mathbf{p2'} = (p1 - p0) * (p2 - p0)$$

$$= \begin{pmatrix} x1 - x0 & x2 - x0 \\ y1 - y0 & y2 - y0 \end{pmatrix} = (x1 - x0)(y2 - y0) - (x2 - x0)(y1 - y0)$$



Line Segments Properties

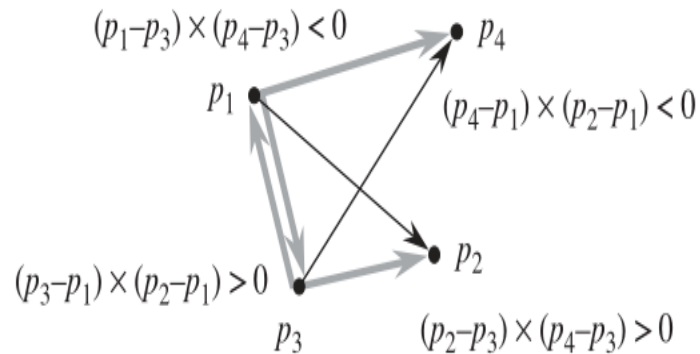


Using the cross product to determine how consecutive line segments $\overrightarrow{p_0p_1}$ and $\overrightarrow{p_1p_2}$ turn at point p_1 .

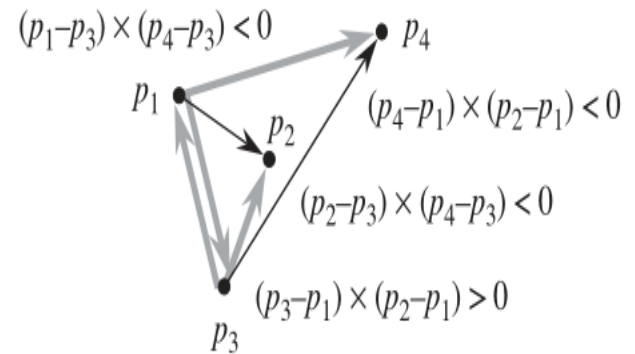
By checking whether the directed segment $\overrightarrow{p_0p_2}$ is clockwise or counter clockwise relative to the directed segment $\overrightarrow{p_0p_1}$.

- (a) If counter clockwise, the points make a left turn.
- (b) If clockwise, they make a right turn.

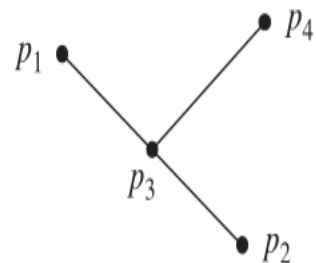
Line Segments Intersection



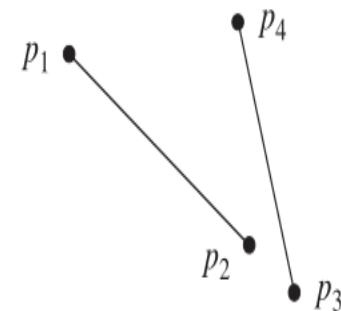
(a)



(b)



(c)



(d)

Line Segments Intersection

SEGMENTS-INTERSECT(p_1, p_2, p_3, p_4)

Returns true if points intersect

```

1   $d_1 = \text{DIRECTION}(p_3, p_4, p_1)$ 
2   $d_2 = \text{DIRECTION}(p_3, p_4, p_2)$ 
3   $d_3 = \text{DIRECTION}(p_1, p_2, p_3)$ 
4   $d_4 = \text{DIRECTION}(p_1, p_2, p_4)$ 
5  if  $((d_1 > 0 \text{ and } d_2 < 0) \text{ or } (d_1 < 0 \text{ and } d_2 > 0)) \text{ and}$   

       $((d_3 > 0 \text{ and } d_4 < 0) \text{ or } (d_3 < 0 \text{ and } d_4 > 0))$ 
6    return TRUE
7  elseif  $d_1 == 0 \text{ and } \text{ON-SEGMENT}(p_3, p_4, p_1)$ 
8    return TRUE
9  elseif  $d_2 == 0 \text{ and } \text{ON-SEGMENT}(p_3, p_4, p_2)$ 
10   return TRUE
11 elseif  $d_3 == 0 \text{ and } \text{ON-SEGMENT}(p_1, p_2, p_3)$ 
12   return TRUE
13 elseif  $d_4 == 0 \text{ and } \text{ON-SEGMENT}(p_1, p_2, p_4)$ 
14   return TRUE
15 else return FALSE

```

Rotation using cross product method

DIRECTION(p_i, p_j, p_k)

```

1  return  $(p_k - p_i) \times (p_j - p_i)$ 

```

Determines whether a point known to be colinear with a segment lies on that segment.

ON-SEGMENT(p_i, p_j, p_k)

```

1  if  $\min(x_i, x_j) \leq x_k \leq \max(x_i, x_j) \text{ and } \min(y_i, y_j) \leq y_k \leq \max(y_i, y_j)$ 
2    return TRUE
3  else return FALSE

```



Thank you

